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Design Evaluation Outcome 3 and Reflection

Test Objective and Validation Metrics

My final Interactive Prototype (IP) 2b tested the three core interactions tested in the previous two IPs, which are: T1 recording with camera by grabbing it, pressing a start and stop button, and placing it on a tripod; T2 selecting a filter by clicking through filter options and selecting one to apply to the whole room; T3 learn a dance, where users do a dance at full speed, 0.5x speed, and full speed again. The IP2b test was done on the Meta Quest headset, instead of on a laptop as had been done for the previous two IPs. The tests aimed to discover whether issues discovered in IP2a and discussed in Design Evaluation Outcome (DEO) 2 were improved, such as: needing help grabbing the camera; not knowing what filters were selected; not knowing whether a bubble had been hit. It also aimed to discover whether users felt they were actually learning the dance through T3 instead of just hitting the targets, which was an uncertain assumption after IP2a.

For each task, success was defined as: a 1-5 ease-of-use rating after each task, and real-time notes during testing, the same as the previous IPs. In response to my IP2a reflection, I also added three questionnaires from the Week 6 lecture post-testing which are: User Experience Questionnaire, Short (UEQ-S) for the overall user experience; Virtual Reality Sickness Questionnaire (VRSQ) for user's simulator sickness; and NASA Task Load Index (NASA-TLX) for how demanding the T3 learn a dance was. In addition, I added 11 post-test questions (PTQ 1-11) to check my improvements after IP2a. These were all done in a Google Form, or during a testing session. Success criteria was set before testing, and can be seen in *Table #1*:

Table #1: Success criteria for IP2b

IP2a finding	IP2b change	Assumption	Success Criteria
3/5 users needed help to grab/attach the camera	Use hands instead of controllers, grab glow, wider snap zone	Beginners can grab and place camera without help	T1 ease above IP2a (min 3.8); PTQ3 grabbing felt natural
Camera not in recording state gets misread	Red button, blinking dot, Stop label and beeps kept	Users still read the recording state	PTQ1 has median of 4 or higher
Unsure which filter was selected; average ease of 4.0	Filter name label, whole-room preview, Apply button	Users can pick a named filter	T2 ease of 4.4 or higher; most users agree with PTQ2
3/5 users couldn't tell if bubbles were hit	Pop sound, colour change, miss state, live score	Bubble hit feedback is clear	PTQ4 median of 4 or higher
Raycast laser aiming is not dancing, 1 user gave up	Bubbles are touched with the hands in VR instead of desktop	Users can hit bubbles with their body	T3 ease above IP2a (min 2.8)
Hitting target bubbles ≠ learning a dance	Practice 0.5x, Full speed, Mastery, attempt history	Users "feel" they learned the dance	Most users agree with PTQ5
Beginners needed help moving, too much walking	'?' hint button at each station, stations closer, Go To panel	Stuck users can find help, moving is easy	PTQ7-8 has a median of 4 or higher

Testing Results

For IP2b, I was able to test with 4 users (U), which is one fewer than planned. Prior to testing, users were asked pre-testing questions in order to have a baseline for T1-3 analysis. Users rated their VR familiarity as 4, 2, 1, and 1 (U1-4 respectively). All users at least somewhat knew the Macarena, the main dance to be tested. U3-4 had previously tested IP1, and only U3 used TikTok daily. Full user pre-testing data is in *Appendix A*.

For U3-4, both their sessions were voice-recorded (see *Appendix D* for transcripts) and their T1-2 was timed. Screen recordings of testing sessions for U2-4 can be seen in my GitHub repository. U1's questionnaire answers were excluded from the analysis, as they did not seem like genuine data. This is because they gave the highest score for each UEQ-S and VRSQ item, and having severe VR symptoms contradicts with high overall VR experience. U1's ease ratings for each task are still shown, but questionnaire results are solely from U2-4. A summary of the data is in *Tables #2-4*:

Table #2: Average ease-of-use (1-5, with 5 being the most easy) across all three IPs

Task	IP1 (5U's, desktop)	IP2a (5U's, desktop)	IP2b (4U's headset)				
			Avg	*U1*	*U2*	*U3*	*U4*
T1 Record with camera	3.3 record, 1.4 tripod	3.8	3.5	5	4	2	3
T2 Selecting a filter	4.4	4.0	4.5	5	5	4	4
T3 Learn a dance	Not built	2.8	4.75	5	5	5	4
T1 time	-	1:41 avg	1:06	-	-	1:42	0:30
T2 time	-	1:23 avg	1:00	-	-	1:38	0:22

Table #3: U2-4 PTQ responses (1-5, with 5 being strongly agree; PTQ success if median ≥ 4)

PTQ 1-11	U2-4 result			Median	# of U's that Agree (result 4-5)	Success Criteria
	U2	*U3*	*U4*			
PTQ1 I could always tell whether the camera was recording	4	4	4	4	3/3	✓
PTQ2 I could tell which filter was selected before applying	4	5	2	4	2/3	✓
PTQ3 Grabbing and placing with my hands felt natural	5	1	2	2	1/3	✗
PTQ4 I could tell straight away when I hit or missed a bubble	3	4	1	3	1/3	✗
PTQ5 I could now do the dance without the bubbles	3	5	5	5	2/3	✓
PTQ6 I felt like I was really inside a dance studio	5	5	2	5	2/3	✓
PTQ7 The ? hint buttons helped me when I was unsure	3	1	4	3	1/3	✗
PTQ8 Getting between the stations was easy	5	5	1	5	2/3	✓
PTQ9 I could read all the text on the panels easily	5	3	1	3	1/3	✗
PTQ10 Brightness changes when switching filters were comfortable	5	4	1	4	2/3	✓
PTQ11 Buttons were easy to reach and press with my hand	5	2	1	2	1/3	✗

Key Findings

T1: The ease rating dropped from 3.8 in IP2a to 3.0 in IP2b, making this the only task to get harder. U2-4 agreed that they were able to tell when the camera was recording, fulfilling PTQ1, but only one user found hand movements in VR (ex. grab, place) natural, which can be seen in the fact that PTQ3 has a median of 2. The 2 least experienced users gave this task their lowest ease

ratings. From the testing, U4 completed it in 0:30, and described the snap to tripod as “very smooth”. U3 took 1:42 and needed help, as the Go To panel was difficult to use, so I moved U3 with the controllers several times.

Table #4: U2-4 questionnaire scores (full scores seen in Appendix B)

Questionnaire measurement & description	U2	U3	U4	Avg
VR familiarity out of 5 (from pre-test questions)	3	2	1	-
UEQ-S pragmatic (-2 to +2) → *“Is it easy to use?” Higher is better*	+2.00	+0.75	0.00	+0.92
UEQ-S hedonic (-2 to +2) → *“Is it fun and interesting?” Higher is better*	+0.75	+2.00	+1.00	+1.25
VRSQ total (0-100) → *“Did it make you feel sick?” Lower is better*	66	25	24	38
NASA-TLX workload (0-5) → *“How hard was the dance?” Lower is better*	0.0	2.0	2.4	1.5
NASA-TLX success at dance (0-5) → *“How well do you think you did?” Higher is better*	5	2	3	3.3

T2: The ease rating went back up to 4.3, which is close to IP1’s rating of 4.4. 2 users said they could tell which filter was selected before applying it, which can be seen in PTQ2 being met. From the testing, U3’s raycast from their hand was “pointing sideways”, so they switched to the VR controllers. Additionally, U3 said “I find the buttons really hard to read, especially the arrows”, and U4 had trouble with the panel, saying “I can't click it… it’s just stuck there”.

T3: The ease rating greatly improved, from 2.8 in IP2a to 4.7 in IP2b. 2 of the users agreed that they could do the Macarena dance without the bubbles, but only 1 was able to tell whether they had hit or missed a bubble immediately. This is reflected in the fact that PTQ5 was met, while PTQ4 was not. Additionally, overall NASA-TLX scores were low, with an average workload of 1.5/5, and physical demand being the lowest at 0.7/5. From the testing, U3 liked the pop sound from hitting the bubbles, and U4 said “I can’t see the balls, I’m missing them” on their initial attempt, then scored 32% and 38% on their later attempts.

Comfort: U2 had a VRSQ total of 66/100, rating eyestrain and headache as ‘Severe’, and other symptoms as ‘Moderate’. U3-4 stayed at an average of 25/100 for VRSQ.

Analysis and Insights

Learn a Dance has been fixed: For T3, ease rating rose from 2.8 to 4.7, which is the biggest improvement in any task across all three IPs. This shows that my two main changes for IP2b were validated: first, touching bubbles with your hands instead of aiming the controller raycast on desktop, which made one IP2a user give up; second, adding a practice mode at 0.5x speed, which an IP2a user asked for. Frustration from the NASA-TLX test was also low with a score of 1.3/5, so learning the dance itself was no longer the source of difficulty.

Hitting targets still may not equal learning the dance: Though 2/3 users said they could now do the Macarena without bubbles, they all partly knew the dance prior. This issue was also raised by IP2a users, who said "maybe [learning the dance is] easier bc i know the song", so PTQ5 may be measuring user’s prior knowledge instead of new dance knowledge that they learn. Users also rated the physical demand to learn a dance to be very low, only 0.7/5 since the bubbles were only for hands. This changes the ‘learning’ into something more like an arm exercise instead of the hand and foot targets I proposed in my initial Design Concept Report (DCR)

Grab-and-place is still the main problem: For all IPs, grabbing and placing objects has been the weakest interaction. The camera’s recording state is no longer an issue, validated by the fact that PTQ1 was 3/3 and had no misreads in IP2a, so the difficulty lies in grabbing the camera and snapping it to the tripod. Previously, users were asking "which button do I press to grab?", but with hand tracking they ask "what hand shape do I make to grab?", thus it was not intuitive and did not remove the problem. This supports my DEO1 finding that grab-and-place has no TikTok equivalent for users to rely on, thus it is not as intuitive for users to interact with.

Prior VR familiarity is a strong distincting factor: For each user, there was a positive correlation between their average PTQ score and their VR familiarity, as the higher their VR familiarity, the higher their PTQ scores were and vice versa. U2-4 had a VR familiarity score of 3, 2, and 1 respectively, and a PTQ average score of 4.3, 3.5, and 2.2 respectively. Of the 11 PTQs, U4 disagreed with 8 of them including text readability, reaching buttons, and moving between stations. The ‘?’ hint buttons that were added for IP2b did not help users, shown in the fact that PTQ7 had a median score of 3, with U3 strongly disagreeing. This is further supported with the UEQ-S test, as hedonitic quality has a score of +1.25, higher than the pragmatic quality of +0.92, which suggests that users find the studio fun before it is easy to use. All this suggests that the lack of VR familiarity causes difficulties with its ease of interaction.

Learn a Dance still needs a clearer indication of feedback: Bubble hit feedback only met its target for 1/3 users (as PTQ4 had a median score of 3). Since the pop and colour change happen at the bubble itself, users may miss it while looking at the example video or straight ahead, which is where a dancer would normally look.

Hand tracking reliability caused much of the friction: The screen recordings show that many problems in T1-2 were mainly caused by hands not being tracked reliably with the VR headset, such as the raycast "pointing sideways", buttons that would not click, and the Go To panel not responding for U3. This explains why U3's T1 took over three times as long as U4's, and why they had to switch to using controllers for T2 and part of T3. Although many of the users did not have much prior experience with hand tracking, this still suggests that it needs larger and clearer targets, a fallback, and maybe even an 'introduction to using hand tracking' module prior to using my VR app before it can replace controllers for beginners.

Comfort must be considered when testing on a headset: U2's VRSQ score of 66 is high for a short session, though they mentioned they were tired from testing other people's VR apps. Since U2 also rated the brightness changes and text as comfortable (both 5), this may be caused by how long the session was or how the headset fit, rather than the filters, though this needs further testing.

Evaluation of Aims

The testing objective was partially met. The results clearly showed that users could complete all three tasks, by recording with the camera, choosing a filter and enjoying learning the dance. However, since my IP2 only had 3 valid questionnaires and limited task data, the test could not fully show whether users actually learned the dance, which was my main question for this IP round.

Table #5: T1-3 Task evaluation

Aim	Result against criteria	Verdict
Users read the recording state	PTQ1 median score of 4, 3/3 users agree	Validated: third round with no misreads
Users pick a named filter	T2 ease rating of 4.3 (though 4.5 with U1), target score was 4.4; PTQ2 median score of 4	Mostly validated
Users hit bubbles with their body	T3 ease rating of 4.7 compared to 2.8 in IP2a	Validated, but accuracy not recorded
Users feel they learned the dance	PTQ5 2/3 users agree, but all knew the song prior, U4 scored only 32-38% in hit accuracy	Uncertain
Moving between stations is easy	PTQ8 median score of 5, but U4 rated it 1/5; Go To panel failed for U3	Partially validated
Hit feedback is clear	PTQ4 median score of 3, target score was 4	Invalidated
Beginners grab and place without help	T1 ease rating dropped to 3.0 from 3.8 in IP2a; PTQ3 median score of 2	Invalidated
Stuck users find help in the ? buttons	PTQ7 median score of 3	Invalidated

Looking across all three IPs, IP2b validated the changes that gave users clear, visible feedback. The T1 camera recording indicator has now gone through all IPs without being misread, and the filter name label brought T2 Filters to its highest ease rating across all IPs of 4.5. T3 Learn a Dance showed the biggest improvement of any task, from an ease rating of 2.8 in IP2a to 4.75 in IP2b, which suggests that changing from desktop with the controller raycast laser to the hand-touch bubbles fixed the key problem from DEO2. This supports the DCR assumption that TikTok-derived user conventions, such as button press affordances, transfer to XR, which has now been held across all three IP rounds.

However, the changes aimed at beginners to VR were mostly invalidated in IP2b. The switch to hand tracking, the grab glow and the '?' hint button at each task did not close the gap between experienced and inexperienced VR users, as U3 and U4 still rated grabbing (PTQ3) and button placement (PTQ11) poorly. This is the third IP where grab-and-place has failed the same assumption. The screen recordings suggest this is partly a technical issue, since U3's hand raycast did not move reliably. Though results may improve if I design my VR app to be more lenient to less 'on target' hand interactions, I think the lack of experience with hand tracking as well as the lack of time for users to go through my testing session may have been a contributing factor to these changes not being validated.

Overall, the main aim of this round, which is whether users actually learn the dance through T3 instead of just hitting the bubbles, still remains uncertain. All users already to some degree knew the Macarena, and U4's in-game scores (32% and 38%) were low despite strong agreement with PTQ5, so user's self-reported learning and actual in-game performance did not match. Thus, the assumption that hitting targets equals learning the choreography is still unvalidated, and though the results of my testing can all point and indicate to some 'direction', they are still inconclusive due to the small number of users for IP2b.

Reflection

IP2b Prototype Session Review

Overall, the parts of IP2b that worked the best were those that were changed in response to earlier IPs. The T1 recording indicator has now gone through all three IPs without being misread, the T2 filter name label brought Filters to its highest ease rating of 4.5, and T3 Learn a Dance became the easiest task once users could touch the bubbles with their hands and use the 0.5x Practice mode. The parts that did not work were those that didn't have a TikTok equivalent, such as: grabbing and placing the camera (PTQ3 with a median score of 2), knowing whether a bubble was hit (PTQ4 with a median score of 3), and reaching small floating buttons (PTQ11 with a median score of 2). This is the same conclusion I came to in DEO1: TikTok users already know how to press buttons, but have no prior knowledge to rely on for spatial affordances and interactions in VR, and beginners like U3-4 felt this the most.

Testing Methodological Reflection Through IP1, IP2a, and IP2b

My testing method improved each IP round. After IP1, I added a VR familiarity rating and a "pass with help" result, and after IP2a, I added hint levels, success criteria set before testing, and the UEQ-S, VRSQ and NASA-TLX questionnaires. However, in practice I only collected part of this plan. The 13-minute session and separate note-taker were hard to keep up in a busy studio, so measures that relied on live notes, such as pass/fail, hint levels, dance accuracy and interview answers, were lost. The recordings also showed that I explained the ease scale backwards to U3 and U4 ("one is easy, five is hard"), and NASA-TLX was answered from 0-5 instead of 0-100, so these results should be read with caution. The main thing I learned is that a few measures collected reliably are more useful than many measures planned. In future, I would use a shorter session, record dance accuracy first since the game already shows it, and test with a room-scale boundary rather than a stationary one.

Overall Concept Evaluation

My original concept of learning and filming TikTok dances in VR was partially confirmed. T1 and T2, which use TikTok-style buttons, were validated across all three IPs, and T3 is now the most enjoyable part of the studio, shown by the UEQ-S hedonic score (+1.25) and presence (PTQ6 median 5). However, the assumption that TikTok conventions transfer to VR did not hold for spatial interactions in any IP, and the assumption that hitting targets equals learning is still unproven. My target audience is also a concern, as the concept is for TikTok users, but the only daily TikTok user (U3) had little VR experience and rated grabbing 1/5.

Improvements and Extensions

- **Keep** the recording indicator, filter name label and whole-room preview, the Go To panel, and the Practice, Full speed and Mastery progression, as these were highly rated and easy to use interactions by users.
- **Remove grab-and-place from T1**: add a "Film me" button that automatically places the camera on the tripod facing the dance spot, and keep grabbing as an option for experienced users, responding to PTQ3 and IP2a.
- **Make learning visible**: Make an actual reflecting mirror of the user, and show hits and misses there, compare each attempt to the previous attempt (ex. "+12% since last try"), and make buttons and text larger in all tasks.
- **Test learning properly**: Use a song users don't already know, record accuracy changes across 3 attempts, and have users stand in a room-scale boundary instead of a stationary one, so they can dance with their whole body.
- **Add T4, scrolling through videos**: Offhandedly and casually, several people mentioned to me that they would like to scroll through videos infinitely like on TikTok. A vertical video feed in the studio, where users swipe to browse and pick a dance to learn, would connect T3 back to how users actually find dances on TikTok.

The concept could be extended using mixed reality (MR) passthrough on the Quest 3, where users dance in their own room and see their real body, with the camera and filters placed on real furniture. This is closer to how TikTok videos are actually filmed and removes the need to walk between stations. Adding body tracking would also create more accuracy with hand tracking, and would bring back the dance foot targets from my DCR.

Hence, an updated concept could look like a TikTok dance studio in MR, where users scroll a feed to find a dance, learn it through bubbles that fade from Practice to Mastery, see their progress after each attempt, apply filters to the whole room, and film themselves with one button without using the tripod.

Appendices

Appendix A: Raw IP2b Data

Appendix A.1 Pre-test answers (Google Form, Week 12)

U#	Date	VR familiarity (1-5)	Used hand tracking	TikTok use *	Filmed a dance	Dance exp. (1-5)	Knows Macarena / YMCA	Tested before
U1	22/10	4	No	Never	No	4	Yes / Yes	No
U2	22/10	3	Not sure	Never	No	1	Yes / Partly	No
U3	27/10	2	Yes	Daily	No	2	Partly / Yes	IP1
U4	27/10	1	Yes	Never	Yes	2	Yes / Yes	IP1

**Note that some users mentioned that they did not use TikTok but I personally know they use Instagram Reels*

Appendix A.2 Ease ratings and NASA-TLX (in-testing note sheets; NASA-TLX answers on 0-5l)

U#	T1 ease	T2 ease	T3 ease	Mental	Physical	Temporal	Success	Effort	Frustration
U1*	5	5	5	1	1	1	5	1	1
U2	4	5	5	0	0	0	5	0	0
U3	2	4	5	4	1	3	2	1	1
U4	3	4	4	2	1	4	3	2	3

Appendix A.3 UEQ-S (5-point form) and VRSQ (0 none, 1 slight, 2 moderate, 3 severe)

U#	UEQ-S items 1 to 8	VRSQ items 1 to 9	Final comment
U1*	5 5 5 5 5 5 5 5	3 3 3 3 3 3 3 3 3	[joke answer]
U2	5 5 5 5 5 4 3 3	2 0 3 2 3 2 2 2 2	Nope
U3	4 4 3 4 5 5 5 5	0 1 1 0 1 2 1 1 0	-
U4	3 3 4 2 4 4 4 4	0 2 2 1 0 1 0 0 0	ymca

Appendix A.4 Prototype statements PTQ1-11 (1 strongly disagree to 5 strongly agree)

U#	PTQ1	PTQ2	PTQ3	PTQ4	PTQ5	PTQ6	PTQ7	PTQ8	PTQ9	PTQ10	PTQ11
U1*	4	4	4	4	4	4	4	4	4	4	4
U2	4	4	5	3	3	5	3	5	5	5	5
U3	4	5	1	4	5	5	1	5	3	4	2
U4	4	2	2	1	5	2	4	1	1	1	1

**Note that U1's responses are straight-lined and excluded from the analysis (see Results section for reasoning).*

Appendix A.5 Testing Questions: Pre-testing, during testing, and post testing:

Appendix A.5.1 Pre-testing questions (Google Form)

1. Who are you (aka name)
2. I agree to take part. I understand my actions and comments will be noted without my name, quotes may appear in the report, and I can stop at any time.
3. How familiar are you with virtual reality (VR)? (1-5)
4. Have you used hand tracking in VR before (using your hands instead of controllers)?
5. How often do you use TikTok?
6. Have you ever filmed a dance video for social media?
7. How much dance experience do you have? (1-5)
8. Do you already know the moves to the Macarena?
9. Do you already know the moves to YMCA?
10. Have you tested an earlier version of this prototype?

Appendix A.5.2 During testing (asked aloud after each task)

1. Ease of use for T1, T2 and T3: “How easy was that, with 1 being the least and 5 being the most?”

Appendix A.5.3 NASA-TLX (asked aloud after T3)

1. Mental: How mentally demanding was learning the dance?
2. Physical: How physically demanding was it?
3. Temporal: How hurried/rushed was the pace?
4. Performance: How successful were you at doing the dance?
5. Effort: How hard did you have to work to get that score?
6. Frustration: How irritated/stressed/discouraged were you?

Appendix A.5.4 UEQ-S (Google Form): “I found the dance studio...” (with a scale of 1-5 between each pair)

1. Obstructive / Supportive
2. Complicated / Easy
3. Inefficient / Efficient
4. Confusing / Clear
5. Boring / Exciting
6. Not interesting / Interesting
7. Conventional / Inventive
8. Usual / Leading edge

Appendix A.5.5 VRSQ (Google Form): “How much is each symptom affecting you right now?” (with a scale of None / Slight / Moderate / Severe)

1. General discomfort
2. Fatigue
3. Eye strain
4. Difficulty focusing
5. Headache
6. Pressure/heaviness of head
7. Blurred vision
8. Dizziness
9. Vertigo

Appendix A.5.6 Post-test questions, PTQ1-11 (Google Form): “How much do you agree with each statement?” (with a scale of Strongly disagree to Strongly agree)

1. I could always tell whether the camera was recording.
2. I could tell which filter was selected before I applied it.
3. Grabbing and placing objects with my hands felt natural.
4. I could tell straight away when I hit or missed a bubble.
5. I could now do the dance without the bubbles.
6. I felt like I was really inside a dance studio.
7. The ? hint buttons helped me when I was unsure.
8. Getting between the stations was easy.
9. I could read all the text on the panels easily.
10. The changes in brightness when I switched filters were comfortable.
11. Buttons were easy to reach and press with my hand.

12. Any final comments?

Appendix B: Scoring and Design Principles

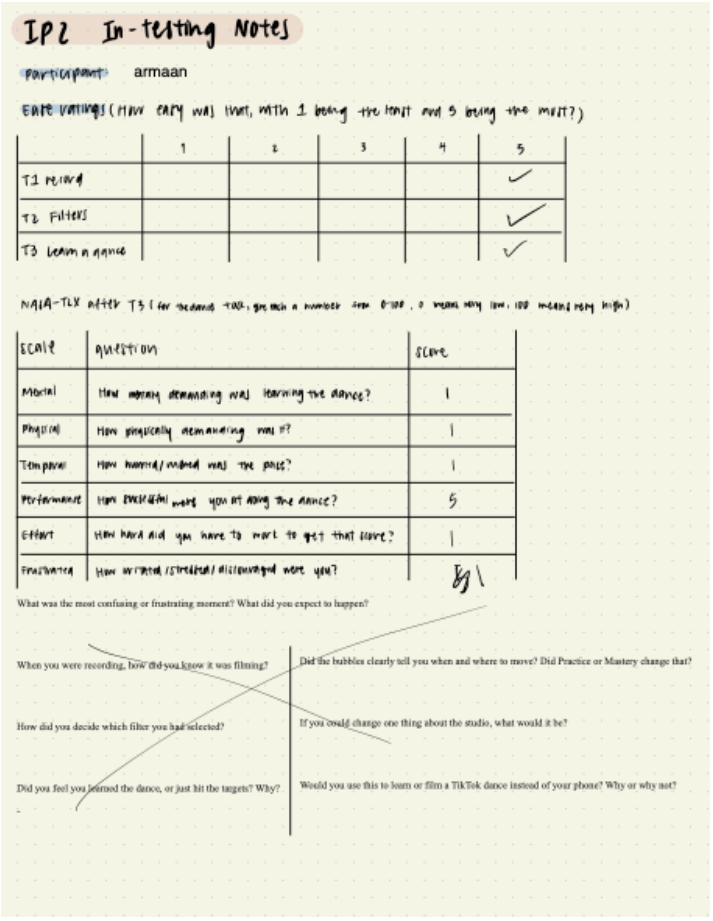
Appendix B.1 Questionnaire Methodologies

- UEQ-S: the form used a 5-point scale, so each answer minus 3 gives -2 to +2. Pragmatic = mean of items 1 to 4 (supportive, easy, efficient, clear); hedonic = mean of items 5 to 8 (exciting, interesting, inventive, leading edge).
- VRSQ (Kim et al., 2018): oculomotor = (sum of items 1 to 4) / 12 x 100; disorientation = (sum of items 5 to 9) / 15 x 100; total = mean of the two.
- NASA-TLX raw (Hart & Staveland, 1988): workload = mean of mental, physical, temporal, effort and frustration. Performance is reported separately because my sheet asked "how successful were you" (higher = better), the reverse of the standard anchor.
- PTQ Likert statements: median and the number of participants answering 4 or 5. IP1 and IP2a values come from Design Evaluation Outcomes 1 and 2.

Appendix B.2 Week 6 design principles against IP2b results (P2 to P4)

Principle	Evidence	Verdict
Text readability	L9 median 3 (5 / 3 / 1); P4 could not read the panels	Partly met: make text bigger and panels closer
Ergonomics	Physical demand 0.7 out of 5; L3 median 2	Dancing is comfortable, grabbing is not
Brightness changes	L10 median 4; P4 rated 1	Mostly met: fade between filters
Avoiding simulator sickness	VRSQ average 38; P2 66 with severe eyestrain	At risk: add breaks, check headset fit
Button placement	L11 median 2 (5 / 2 / 1)	Not met: buttons hard to reach for 2 of 3

Appendix C: In-testing Note Sheets



IP2 In-testing Notes

Participant armaan

Ease ratings (How easy was that, with 1 being the least and 5 being the most?)

	1	2	3	4	5
T1 reload					✓
T2 Filters					✓
T3 Learn a dance					✓

NASA-TLX after T3 (for seconds too: give each a number from 0-100, 0 means very low, 100 means very high)

Scale	Question	Score
Mental	How mentally demanding was learning the dance?	1
Physical	How physically demanding was it?	1
Temporal	How hurried/rushed was the pace?	1
Performance	How successful were you at doing the dance?	5
Effort	How hard did you have to work to get that score?	1
Frustration	How irritated/stressed/discouraged were you?	1

What was the most confusing or frustrating moment? What did you expect to happen?

When you were recording, how did you know it was filming?

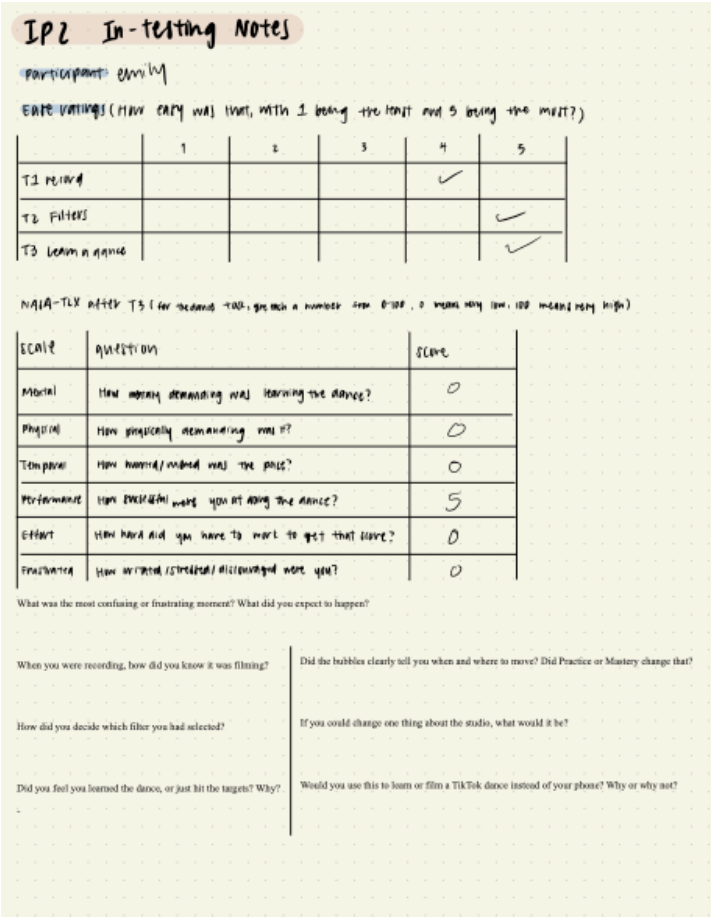
Did the bubbles clearly tell you when and where to move? Did Practice or Mastery change that?

How did you decide which filter you had selected?

If you could change one thing about the studio, what would it be?

Did you feel you learned the dance, or just hit the targets? Why?

Would you use this to learn or film a TikTok dance instead of your phone? Why or why not?



IP2 In-testing Notes

Participant emily

Ease ratings (How easy was that, with 1 being the least and 5 being the most?)

	1	2	3	4	5
T1 reload				✓	
T2 Filters					✓
T3 Learn a dance					✓

NASA-TLX after T3 (for seconds too: give each a number from 0-100, 0 means very low, 100 means very high)

Scale	Question	Score
Mental	How mentally demanding was learning the dance?	0
Physical	How physically demanding was it?	0
Temporal	How hurried/rushed was the pace?	0
Performance	How successful were you at doing the dance?	5
Effort	How hard did you have to work to get that score?	0
Frustration	How irritated/stressed/discouraged were you?	0

What was the most confusing or frustrating moment? What did you expect to happen?

When you were recording, how did you know it was filming?

Did the bubbles clearly tell you when and where to move? Did Practice or Mastery change that?

How did you decide which filter you had selected?

If you could change one thing about the studio, what would it be?

Did you feel you learned the dance, or just hit the targets? Why?

Would you use this to learn or film a TikTok dance instead of your phone? Why or why not?



IP2 In-testing Notes

Participant: Steph

Ease ratings (How easy was that, with 1 being the least and 5 being the most?)

	1	2	3	4	5
T1 record		X			
T2 Filters				X	
T3 Learn a dance					X

NASA-TLX after T3 (for the next two, give each a number from 0-100, 0 means very low, 100 means very high)

Scale	Question	Score
Mental	How mentally demanding was learning the dance?	4
Physical	How physically demanding was it?	1
Temporal	How hurried/rushed was the pace?	3
Performance	How successful were you at doing the dance?	2
Effort	How hard did you have to work to get that score?	1
Frustration	How irritated/stressed/discouraged were you?	1

What was the most confusing or frustrating moment? What did you expect to happen?

When you were recording, how did you know it was filming?

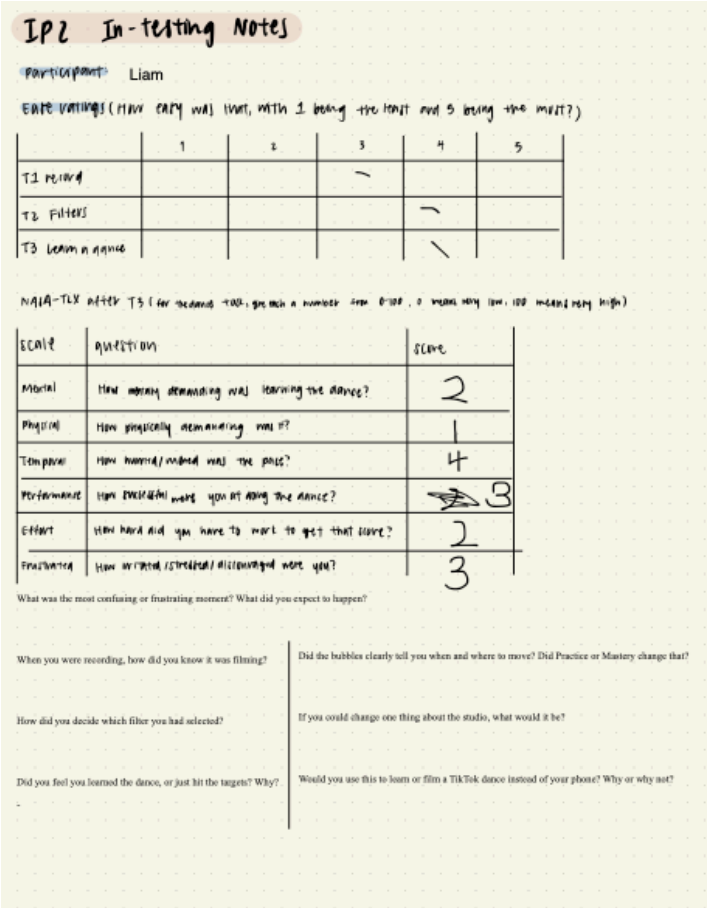
Did the bubbles clearly tell you when and where to move? Did Practice or Mastery change that?

How did you decide which filter you had selected?

If you could change one thing about the studio, what would it be?

Did you feel you learned the dance, or just hit the targets? Why?

Would you use this to learn or film a TikTok dance instead of your phone? Why or why not?



IP2 In-testing Notes

Participant: Liam

Ease ratings (How easy was that, with 1 being the least and 5 being the most?)

	1	2	3	4	5
T1 record			~		
T2 Filters				~	
T3 Learn a dance				\	

NASA-TLX after T3 (for the next two, give each a number from 0-100, 0 means very low, 100 means very high)

Scale	Question	Score
Mental	How mentally demanding was learning the dance?	2
Physical	How physically demanding was it?	1
Temporal	How hurried/rushed was the pace?	4
Performance	How successful were you at doing the dance?	~~3~~
Effort	How hard did you have to work to get that score?	2
Frustration	How irritated/stressed/discouraged were you?	3

What was the most confusing or frustrating moment? What did you expect to happen?

When you were recording, how did you know it was filming?

Did the bubbles clearly tell you when and where to move? Did Practice or Mastery change that?

How did you decide which filter you had selected?

If you could change one thing about the studio, what would it be?

Did you feel you learned the dance, or just hit the targets? Why?

Would you use this to learn or film a TikTok dance instead of your phone? Why or why not?

Appendix D: Session Transcripts (U3, U4)

U3-4's testing sessions were recorded with consent, and the voice recordings were transcribed using Otter AI (. Some words may be transcribed incorrectly.

Screen recordings of the sessions for U2-4 were done using Meta Quest Developer Hub Casting, and the sessions are available in my GitHub repository.

Appendix D.1 U3 transcript

Speaker 1 0:00
You consent to be recorded. Sorry. Okay. And can I just drive

Speaker 2 0:12
to stand or something?

Speaker 1 0:13
Nah. You do you wanna?

Speaker 2 0:16
No.

Speaker 1 0:17
Okay. Don't. It's fully handled.

Speaker 3 0:33
So if you turn, oh no, like.

Speaker 1 0:46
Are you okay, my dear? Okay. Also, if at any point you aren't able to move to the place you want to move to, let me know, and then I'll use the controllers to move you. Okay. So on your left there is a welcome screen. Okay, so try pressing. So to grab is like a normal grab, and then to to press is like a pinch. So if you want to press the button, make the ray cast go on the button, and then pinch. Okay, so let's do task number one. So I I want you to oh if you see on the right side there's a panel. So that's kind of your jump to the different no no no no no too far. Yeah that one that sorry left left

Speaker 2 1:38
this

Speaker 1 1:39
one yeah oh that panel is your jump to different locations button. So I want you to go to the camera and then record.

Speaker 2 1:53
Press record now.

Speaker 1 1:54
Okay. So if you press record, it should. Okay. Oh no, it's not. You're not jumping to the record area. It should work. Why isn't working? This

Speaker 4 2:10
sucks.

Speaker 1 2:11
It's okay if it doesn't work. It's a bit of a finicky guy. Okay, that's fine. I will. I will just manually move you. I'm gonna use the controller and move you. Just let you know. Why? Why isn't it? What the fuck? You know. If anything, hey Liam. Liam is my. Oh, here we go. Okay. So as you can see, here are keeping it. Here are the. Here is the. So if you can grab. Wait, Oshay, I need some time. I'm gonna time this task so at least I have one thing that's timed. Okay, you're gonna grab the camera and press the record button, and then after a few seconds, take off the record button.

Speaker 2 3:21
Oh, sorry. have to? Oh, okay. And

Speaker 1 3:25
then it's kinematic, so it won't fall. Yeah. So what we do is then press the record button. Do the do the pinchy. It's because one you don't you don't have to grab it if you're left.

Speaker 2 3:36
I'm confused.

Speaker 1 3:37
Yeah. Just just. Okay. Yep. And then you can stop recording whenever you want. Okay. And then, if you can, on the right side, do you see? There's a tripod over there.

Speaker 2 3:55
Yeah.

Speaker 1 3:55
Yeah. So put the camera on the tripod. It will snap into place.

Speaker 2 4:03
How do I get there? Oh, oh, wait. No, I don't know. I don't know. Give me, give me.

Speaker 1 4:09
Yeah. Okay, I'm moving you. Okay, so I really need to work on this move. Can you, can you, can you grab it while you're doing that? Oh shit! Okay, grab it. Let's grab it. Okay, okay. All right, come on. It's a team effort. You gotta. Oh, it disconnected. Why did it disconnect? How was it going?

Speaker 2 4:44
getting really hot.

Speaker 1 4:45
You're getting really hot.

Speaker 2 4:46
Yeah.

Speaker 1 4:47
Okay, but why did the casting stop? We can take a pause. I

Speaker 2 4:51
don't know what's happening.

Speaker 1 4:52
I don't know what's happening either. How did? Why

Speaker 2 4:53
did it come

Speaker 1 4:55
out? I don't know. You're the you're the Unity guy. Is it

Speaker 5 4:57
plug in?

Speaker 2 4:58
Yeah.

Speaker 1 5:01
Okay, maybe I stop casting.

Natasha Lim 5:05
It's not even connected.

Speaker 1 5:06
It's not connected. Wait, no, no, no, no, no! You can't. You're not allowed to. I'm gonna stop you. Yeah, here you can take it off. It disconnected.

Speaker 2 5:18
Oh,

Speaker 1 5:18
I don't know what's happening. Okay. Lim, what did you do?

Natasha Lim 5:31
I didn't do nothing.

Speaker 1 5:33
This is all your fault. Wait. Huh? How did this even happen? I'm so puzzled. Why does it even? Why did it even switch to a 3s? What is happening? Device detected. I totally could have. It was pretty good so far, right? You saw that. You saw that.

Natasha Lim 6:05
Yeah, it was working. Looked great.

Speaker 3 6:07
I know.

Speaker 1 6:11
Okay.

Speaker 3 6:20
But you also pushed it a bit, by the way. Okay. Okay, it's working again. I'll just

Speaker 1 6:33
wait for her, I guess. I

Speaker 3 6:36
told you. Yeah, but what's the yeah? Make sure you yeah pretty much it's now the.

Speaker 3 6:43
those missing scripts.

Speaker 4 6:45
Yeah. Pleasure. Can you pause the recording?

Speaker 1 6:55
Yeah, you can. 7%. Holy battery.

Speaker 3 6:59
So now it's now it's the time to see what needs to be completed.

Speaker 1 7:03
Okay, unpaused, and now let me move you. Okay, the camera I moved it, so it should be. Sorry, you must. Hopefully, you don't get too dizzy, my dear.

Speaker 2 7:19
Oh, don't.

Speaker 1 7:21
Oh, okay. I will position you

Natasha Lim 7:23
right there. Why didn't this look at that zoom thing?

Speaker 1 7:28
Oh, it's kind of weird. Okay. Oops. Is that is that okay? You can see the tripod is there and the camera is there. Yeah. Okay. I'm gonna continue the timer. So put the camera on the tripod.

Speaker 2 7:41
Okay, now, like when am I going?

Speaker 1 7:44
No, no, no, no. Yeah, any time, tomorrow.

Speaker 6 7:50
Done.

Speaker 1 7:51
Yay! Okay, done. Task one. So, how easy was? Oh, how easy was that on a scale of one to five?

Speaker 2 8:05
So

Speaker 1 8:07
five is very five is one is easy, five is hard.

Speaker 2 8:12
Am I factoring in all the things that went wrong?

Speaker 1 8:15
Yeah, yeah, just in general too. Okay, damn.

Speaker 2 8:18
Sorry,

Speaker 1 8:19
that's okay. Okay, then I want you to go to. There's a mirror filter thing.

Speaker 7 8:30
Yeah.

Speaker 1 8:31
You can wait. Try the try the go to button. It should work. Damn. Why isn't it working? That sucks. Do you want me to move you there then? Okay. Sorry if sorry if dizzy. Okay, so there is a mirror button here. So if you can press the arrows and switch to the star filter and apply it, and let me now.

Speaker 2 9:05
My red cross is like pointing sideways. You know, see.

Speaker 1 9:10
I see.

Speaker 2 9:11
Okay, I'll just use my left. Oh, the star. Oh,

Speaker 1 9:17
if it's if it's ever too hard, you can try the controllers, but it should all work with hands. Do you want me to move you closer?

Speaker 7 9:29
Maybe.

Speaker 1 9:36
Okay, this is close as I can get you, my dear. Okay. Oh, holy six of them! I don't know, my dear. Oh, and we're back. And we're back to the stars. Who the water? Oh. The button sock, man. Oh, that's not my fault. I look at you like you have the solution.

Speaker 8 10:20
What's going wrong?

Speaker 3 10:22
Everything.

Speaker 1 10:22
Do you want to use the controller? That might be easier. Here, you only need one. This is your yeah. This is your left one. You can. Okay. Oh, and then apply. Okay. Done. Stop. So, how easy was that on a scale of one to five?

Speaker 2 10:47
I guess four. The buttons were

Speaker 1 10:50
four.

Speaker 2 10:51
Um, I find the buttons really hard to read.

Speaker 1 10:54
Okay.

Speaker 2 10:55
Especially the arrows.

Speaker 1 10:56
Okay, that's good advice. You are the best. Now I need you to go to the dance area. You can navigate yourself now. Here, I did it. Yeah. Okay. So this one, this one, you actually have to use just hands, though. Oh my

Speaker 2 11:15
god, I'm scared.

Speaker 1 11:17
So on the, you need to take a break.

Speaker 2 11:20
Pardon?

Speaker 1 11:20
You can take a break. Okay. Okay. On the right side, there is a button. Yes. So, if you do, you want to do Macarena or YMCA?

Speaker 2 11:29
Let's do YMCA.

Speaker 1 11:30
Okay. So if you recast and on to YMCA. Hello. Damn, bro! Why does it like hate you? Sorry, I didn't mean it. Here, maybe, maybe. Oh, I guess

Speaker 2 11:53
we stop between the macarena.

Speaker 1 11:55
No, no, no! Here, let me, let me.

Speaker 2 11:57
Oh, I did it. Okay,

Speaker 1 11:58
okay, okay, and then. Oh, here, let me. Here, controller. And then you press full speed, and then look straight. What am I doing?

Speaker 2 12:23
I can't believe this.

Speaker 1 12:31
Look, there's just dance on the side.

Speaker 2 12:56
No, I really like that.

Speaker 1 12:57
You liked it? Yeah.

Speaker 2 13:01
It's just like all the other sounds previously were just like the buttons clicking, right? It's so stupid, but really good.

Speaker 1 13:16
It was so mellow as well before this class. Put the camera on the heel. Guys, I aim to please. Surprise, motherfuckers! Can you do it again? Yeah, yeah. Hey, you can do the macarena.

Natasha Lim 13:31
What do the bubbles actually do? They actually. And you miss me. So if

Speaker 1 13:34
you hit it, it'll be green. If you don't hit it, it'll turn red. Yeah.

Natasha Lim 13:39
Why are they all purple?

Speaker 2 13:41
Because I didn't hit the meal, I guess. You

Speaker 1 13:42
didn't hit them yet.

Speaker 2 13:44
Okay.

Speaker 1 13:44
You can try again. You do macarena.

Speaker 2 13:50
Can I have control, please?

Speaker 1 13:51
Oh yes, yes, yes.

Speaker 2 13:52
And I'm out. You're

Speaker 1 13:53
out.

Speaker 2 13:54
I just see you guys. This one. I just see you guys.

Speaker 1 14:02
How do I fix it for you, my dear?

Speaker 6 14:03
I don't know.

Speaker 1 14:05
Do you want to take it off? Yeah, me ripping it off your face. Oh, I see it though. Oh. Here, I'll do the macarena.

Speaker 2 14:14
You did it.

Speaker 1 14:17
Where'd it go? Where's my? Where's my? Oh. Oh, do

Speaker 2 14:23
you want? No, no,

Speaker 1 14:24
no. It should be okay. I need it. I need it to here. Get put the. I'm trying

Speaker 2 14:30
to hide it. It's out of bounds.

Speaker 1 14:31
Oh, oh. Take it. Look at my hands, bro.

Speaker 9 14:42
Okay.

Speaker 1 14:47
Okay, Macarena is selected. See? Oh my gosh, it's so cool! Then it turns green. Yeah, guys, I know it's those fuck. Then there's more bubbles. I don't know. This is a later fall. You can honestly just do this and pass

Speaker 10 15:20
it. Yeah.

Speaker 1 15:20
Anyways, look at that, and then this is the real YMCA now. If it loads,

Speaker 2 15:32
oh!

Speaker 1 15:40
Isn't it beautiful? I had to edit it, you know. Oh. What do you think?

Speaker 3 15:54
If you did, what do you think? Yeah. You want to try it again?

Speaker 1 16:00
No more. Okay. Okay. Anyways, oh, I have some more questions for. Okay. But I need to find them because they're on my iPad and I don't have my iPad with me today. Okay. How mentally demanding was learning the dance. One to five.

Speaker 2 16:34
Same, same as just now. Ignore this. So five being it wasn't really demanding. Five

Speaker 1 16:39
being it was extremely.

Speaker 2 16:40
Oh, learning the dance.

Speaker 1 16:44
Yeah. Oh, like

Speaker 3 16:46
one. One. Okay. Wait, you said

Speaker 1 16:48
4040, how how easy was the dance?

Speaker 2 16:58
455.

Speaker 1 17:02
How physically demanding was learning the dance?

Speaker 1 17:05
Out of five. How hard, rushed was the pace?

Speaker 2 17:14
I think the transition from full speed to dance that was a bit fast. So maybe like three.

Speaker 1 17:23
How successful were you at doing the dance? You got um because it was

Speaker 3 17:30
two. I think you did great though.

Speaker 1 17:33
How hard did you have to work to get that score? Not very hard. One. How irritated, stressed,

Speaker 11 17:39
discouraged were you? I wouldn't

Speaker 2 17:41
say irritated or distressed. Okay, then I have some post

Speaker 1 17:45
questions. Sorry, so many questions. No, no, no, not everything. Here, all of them is kind of just stupid. Wait, so if I if I if I if I stop app and launch app, it'll come back again. Correct.

Speaker 7 18:01
Yeah, and

Speaker 1 18:02
then I'll write

Speaker 3 18:03
it. Yes, but the scripts have you not kept them

Speaker 1 18:07
in

Speaker 3 18:11
a separate folder?

Speaker 1 18:16
Okay, okay, okay. Oh no, not

Speaker 3 18:19

that folder, but the other assets for that canvas. Oh, me and my friend have decided that when I'm going back to Singapore, me and her we get

Speaker 1 18:30
to have one more fun silly time in Singapore in Singapore. Yeah. What is that so wrong? I mean, it

Speaker 3 18:41
works. See, I don't feel. So that's the reason why I'm being nitpicky about that stuff. It helps though when you are nitpicky about that kind of stuff. It doesn't feel like I

Unknown Speaker 18:55
don't really know why. End recording. Bye.

Appendix D.2 U4 transcript
Speaker 1 0:00
Hands

Speaker 2 0:01
are big.

Speaker 1 0:02
It's like close.

Speaker 3 0:03
You consent to being recorded. Thanks. Okay, so let's. So this is the thing for the in this prototype. Grabbing is like a like a like a grab, and then pointing is like a pinch with a ray cast. So let's try with the let's go. Okay, so now task number one. I want you to go try with the oh on the right side you see there's a go to panel, so that's kind of your jump around panel. So you press record. Yes, and then the task number one is you're gonna grab the camera and then press record and then put it on the tripod 321. You don't have to, yeah. And then you can you can stop the recording. It's kinematic, yeah. It won't it won't fall. Okay,

Speaker 1 1:09
put it again.

Speaker 3 1:14
Yeah, and then put it on the tripod, and let me know if you need me to move you. Is

Speaker 1 1:20
there a

Speaker 3 1:22
maybe try going to

Speaker 4 1:23
stop?

Speaker 3 1:24
Oh, stunning! You see how smooth that throw is. That was very

Speaker 5 1:29

smooth. All right, how

Speaker 3 1:30
hard is that on a scale of one to five?

Speaker 1 1:34
Five is the hardest. Five is the

Speaker 3 1:35
hardest.

Speaker 1 1:38
Three.

Speaker 3 1:39
Three. Damn, it'd be like that. Oh my gosh, my keyboard. Okay, whatever. Um, let's go to task number two, which is filters. So as you can see, right in front of you, there is a nearest station. I don't know why is that right in front. Anyways, I want you to put it into

Speaker 6 2:02
stop. Can't

Speaker 1 2:03
move this guy away from

Speaker 3 2:05
Rio de Janeiro.

Speaker 1 2:07
I can't click it. It's like ocean. Oh, I can't wait.

Speaker 7 2:14
It

Speaker 3 2:16
can't. It's it's just stuck there. I got it.

Speaker 1 2:20
I got it.

Speaker 3 2:22
I don't know how to. That's a that's a using error. I'm

Speaker 1 2:25
gonna like.

Speaker 3 2:26
Okay, click it to Rio de Janeiro Rio filter. I

Speaker 1 2:30
have no idea what is original. Oh, you'll see.

Speaker 3 2:35
Rio de Janeiro. You can you can look at this or how about water water filter. Okay.

Speaker 1 2:44
No, I have another one.

Speaker 3 2:46
And then press apply. I

Speaker 1 2:48
want. I like this

Speaker 3 2:49
one. Okay. Then press apply. Okay. And how hard was that on a scale of one to 544. Okay. That

Speaker 1 3:01
was like in front of me.

Speaker 3 3:02
Okay. Now go to the learner dots.

Speaker 8 3:05
Oh.

Speaker 3 3:08
No. Do you want to do the Macarena or do you want to do? Wait.

Speaker 1 3:12
Can I get back? I don't like this filter. Oh. Okay. Sorry, it was too bright for me.

Speaker 3 3:19
Damn. I was concerned about the brightness. Um, you wanna learn the macarena or the YMCA? You pinch, pinch, same, same. Suck. And then fast speed. Look straight. I can't see the voles. Wait, I'm missing them.

Speaker 1 4:06
But never get those ones.

Speaker 3 4:14
Well, at least you're hearing some. Okay, you want to try YMCA?

Speaker 1 4:20
Yes, please.

Speaker 3 4:22
What if I said no?

Speaker 1 4:24
That's it. That's the end of the test. I'm on mastery. Hang on. Oh shit! I can't even see them.

Speaker 3 4:30

The bubbles disappear fast. They're on mastery mode. Do you like the just dots in the corner?

Speaker 1 4:41
Yeah, I almost didn't notice it. Oh, hang on already. I can't see the bubbles.

Speaker 3 4:46
Yeah, because you're on last stream mode, they disappear faster.

Speaker 9 4:51
32%.

Speaker 1 4:53
Shit, that's kind of terrible. Oh.

Speaker 3 4:57
Anyways, that's that's all. What does earlier and later do? Earlier in the song and later in the song. If you wanted to go back and practice one specific mood. Okay. Well, maybe it doesn't work. No one's tested up till now. Me too. Okay. Yippee! Yippee!

Speaker 1 5:21
Thank you.

Speaker 3 5:22
Okay, I'm not done. You have to, you have to, you have to. They

Speaker 1 5:27
said bright.

Speaker 3 5:28
I know. I'm sorry. I don't know why that happened for you. Why is this not working, guys? Yeah, exactly. I feel like you will know. Why is my keyboard not working, honey? Why is my super soup? I don't know

Speaker 1 5:41
what the fuck.

Speaker 3 5:43
How mentally demanding was learning the dance? Not a five. Okay,

Speaker 1 5:49
sorry. Two.

Speaker 3 5:53
Damn, my keyboard is not working.

Speaker 1 5:56
Is your font too small or too big?

Speaker 3 5:58
No, I'm the best. Okay, you know what? Wait, record was three. Filters was four. How was how was how easy was learning to dance out of 520,

Speaker 1 6:18
easy

Speaker 4 6:19
four. And then, how mentally demanding?

Speaker 3 6:23
Two. Two. How physically demanding?

Speaker 10 6:26
One. How hard slash

Speaker 3 6:30
rush was the pace? Like the

Speaker 8 6:34
bubbles came.

Speaker 3 6:36
How successful were you at doing the

Speaker 11 6:38
dance? What's

Speaker 1 6:39
38%? Two.

Speaker 3 6:41
You did three. That's a 312. You got it. Three. How hard did you have to work to get that score? Two. Are you judging my dancing more than judging the discouraged were you?

Speaker 1 7:06
Oh no, like a three.

Speaker 3 7:09
Damn, you were discouraged, bro. Skill issue. Sorry, didn't mean it. Okay.

Speaker 1 7:18
I found

Speaker 3 7:20
it at studio.

Speaker 1 7:22
Oh. Oh. I don't think it's either of these.

Speaker 3 7:31
There's a lot of them. Don't worry. Okay. Okay. You just. Okay. End recording. Bye bye.